INCH-POUND

MIL-M-63321A (AR)
AMENDMENT 4
30 APRIL 1990
SUPERSEDING
AMENDMENT 3
8 JULY 1986

MILITARY SPECIFICATION

MICROCIRCUIT, DIGITAL, CMOS (FINAL LOGIC)

This Amendment forms a part of Military Specification MIL-M-63321A (AR) dated 8 April 1981, and is approved for use within the US Army Armament Munition and Chemical Command and is available for use by all the Departments and Agencies of the Department of Defense.

PAGE 2

* 2.1 Drawings, Delete “9292963” and substitute "12927056 and 9292963".

PAGE 3

3.4 Mechanical shock, Conditions: Delete "D" and substitute "C".

Delete "Method 2105.1” and substitute "Method 2015".

PAGE 5

3.5.4., (a) and (b): Delete (-45°C to +85°C) and substitute (-40°C to +70°C).

PAGE 6

* 3.6.2 Delete “9292963x..., as applicable” and substitute "12927056X or 9292963X where "X" denotes the drawing revision letter, as applicable".

PAGE 8

4.3.1 Submission: Delete last sentence and substitute “The first article sample, after screening, shall consist of the quantities needed for double samplings stipulated in Table III".

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:38Z
Check the source to verify that this is the current version before use.

4.3.4 Add "e. A lapse in production for a period exceeding 90 days or as stipulated in the contract".

PAGE 9

* TABLE III, Drawing number, Delete “9292963” and substitute “12927056 and 9292963”.

TABLE III, Subgroups a, b, c, d: Delete “25” and substitute “note 8".

Subgroup b, Mechanical shock: Delete condition "D" and substitute condition "C".

Subgroup c, Resistance to solvent: Delete "Method 2015.1" and substitute "Method 2015".

PAGE 10

TABLE III, Subgroup g: Delete "25" and substitute "Note 9".

NOTES: Delete present content of note 1 and substitute "Devices which fail any test criteria in the screening sequence shall be removed from the first article sample. The contractor should choose a first article sample that will provide, after screening, the quantities stipulated in notes 8 and 9.

NOTES : Delete content of note 8 and substitute:

A double sampling plan shall be used as follows:

25-0-2, 38-1-2. For group b replace fallouts from subgroup a.

Add “note 9. A double sampling plan shall be used as follows:

45-0-2, 77-1-2.

PAGE 12

4.4.2.1 Drawing number Delete “9292963” and substitute “12927056 and 9292963"

PAGE 13

4.4.2.2 Drawing number, delete “9292963” and substitute “12927056 and 9292963”.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:38Z
Check the source to verify that this is the current version before use.

4.4.2.2 AQL or 100% Column: Delete "All LTPD's" and substitute "LTPD's of 10%, 10%, 10% and 10%".

PAGE 14

4.4.2.3 Subgroup 2, Resistance to solvents: Delete "Method 2015.1" and substitute "Method 2015".

Delete "Bond strength (bond pulls 1/), ---, LTPD 15%". and substitute "Bond strength (bond pull 1/, 5, 1-2".

Delete Notes "1/" in its entirety and substitute: "1/ Pull all bond wires on each device - reject for two(2) or more bond failures".

* 4.4.2.3 Drawing number, Delete "9292963" and substitute "12927056 and 9292963".

PAGE 15

* 4.4.2.4 Drawing number, Delete "9292963 and substitute "12927056 and 9292963".

4.4.2.4 Subgroup 2, Mechanical shock: change Condition "D" to Condition "C".

PAGE 19

4.5.16 Burn-in screen Line 6: Delete "7.5 volts DC" and substitute "8.0 volts DC".

4.5.17 Steady state life line 3: Change "Test duration shall be 100 hours minimum at 125°C + 5°C." to "Test duration shall be 1000 hours minimum at 125°C + 5°C or equivalent per Method 1005 of MIL-STD-883B. Temperature in excess of 135°C + 8°C and test duration less than 496 hours shall not be allowed".

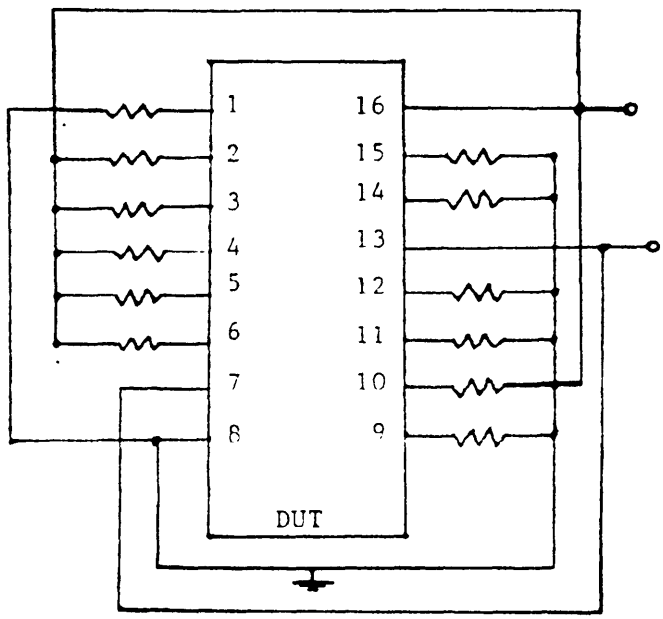
PAGE 20

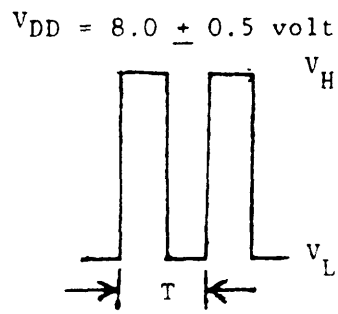
6.3 and 6.5 Delete "QAA" and substitute "QAT".

PAGE 30

FIGURE 11: Delete Figure 11 and substitute revised FIGURE 11 included in this amendment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:38Z
Check the source to verify that this is the current version before use.





V_H = 8.0 - .5 volts

V_L = 0.5 ± 0.15 volts

T = 40 ± 4 µs period, 50% duty factor.

All (Rs) = 51K ohms max.

FIGURE 11. Burn-in and steady state of life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:38Z
Check the source to verify that this is the current version before use.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army-AR

Preparing activity
Army-AR

(Project 1345-A567)

Source: https://assist.dla.mil -- Downloaded: 2026-04-23T02:38Z
Check the source to verify that this is the current version before use.